How to Execute Task 2: spaCy One Row Per Token

This experiment analyzes a sentence at multiple NLP levels using **spaCy**.

Step 1: Install spaCy

Open Command Prompt (Windows) or Terminal (Linux/Mac):

```
pip install spacy
```

Step 2: Download the English Language Model

Run:

```
python -m spacy download en_core_web_sm
```

This downloads the language model required by the code.

Step 3: Create the Python Program

Save the following as spacy_token_analysis.py:

```
import spacy

nlp = spacy.load("en_core_web_sm")

doc = nlp(
    "The children in Pune didn't eat "
    "the mangoes because they were raw."
)

 print(f"{'TOKEN':<12} {'LEMMA':<12} {'MORPH':<25} {'POS':<8} {'DEP':<10} {'HEAD'}")
 print("-" * 80)

 for t in doc:
     print(
         f"{t.text:<12} "
         f"{t.lemma_:<12} "
         f"{str(t.morph):<25} "
         f"{t.pos_:<8} "
         f"{t.dep_:<10} "
```

```
        f"{t.head.text}"
    )

print("\nEntities:")
for ent in doc.ents:
    print(ent.text, ent.label_)
```

Step 4: Run the Program

```
python spacy_token_analysis.py
```

Expected Output

```
TOKEN        LEMMA       MORPH                    POS      DEP
HEAD
------------------------------------------------------------------
-------------
The          the         Definite=Def             DET      det
children
children     child       Number=Plur              NOUN     nsubj
eat
in           in                                   ADP      prep
children
Pune         Pune        Number=Sing              PROPN    pobj
in
did          do          Tense=Past               AUX      aux
eat
n't          not         Polarity=Neg             PART     neg
eat
eat          eat         VerbForm=Inf             VERB     ROOT
eat
the          the         Definite=Def             DET      det
mangoes
mangoes      mango       Number=Plur              NOUN     dobj
eat
because      because                              SCONJ
mark       were
they         they        Number=Plur              PRON
nsubj      were
were         be          Tense=Past               AUX
advcl      eat
```

```
15  raw          raw          Degree=Pos                ADJ
acomp      were
```

```
1  Entities:
2  Pune GPE
```

Understanding Each Column

Column	NLP Layer	Meaning
TOKEN	Lexical	Original word
LEMMA	Morphology	Base form of word
MORPH	Morphology	Grammatical features
POS	Syntax	Part of Speech
DEP	Syntax	Dependency relationship
HEAD	Syntax	Parent word in dependency tree
Entities	Semantics	Named entities

Morphological Analysis

Examples:

```
1  children → child
2  mangoes → mango
3  were → be
```

Morphology studies how words change form.

Syntactic Analysis

From the output:

```
1  children  nsubj  eat
2  mangoes   dobj   eat
```

This means:

- **children** = subject
- **mangoes** = object
- **eat** = main verb

Dependency tree:

```
1  eat
2  ├── children
3  ├── mangoes
```

```
└── were
    └── raw
```

Semantic Analysis

Named Entity Recognition identifies:

```
Pune → GPE (Geo-Political Entity)
```

Meaning representation:

```
EAT(children, mangoes)
LOCATION(Pune)
CAUSE(raw(mangoes))
```

Discourse Question

Which word does "they" refer to?

Sentence:

```
The children in Pune didn't eat the mangoes because they were raw.
```

Answer:

```
they = mangoes
```

because **raw** naturally describes **mangoes**, not children.

Viva Questions

1. What is a lemma?

The dictionary or base form of a word.

Example:

```
children → child
running → run
```

2. What does POS mean?

Part of Speech.

Examples:

```
NOUN
```

```
VERB
ADJ
PRON
```

3. What is dependency parsing?

Finding grammatical relationships between words.

Example:

```
children → subject of → eat
```

4. What is a named entity?

A real-world entity such as:

```
Person
Location
Organization
Date
```

Example:

```
Pune → Location (GPE)
```

5. What NLP layers are demonstrated here?

- Morphology
- Syntax
- Semantics
- Pragmatics
- Discourse

Conclusion

This experiment demonstrates how **spaCy** analyzes a sentence token by token to extract:

- **Morphological information** (lemma, grammatical features)
- **Syntactic information** (POS tags, dependency relations)
- **Semantic information** (named entities)
- **Discourse-level understanding** (pronoun reference)

It shows how a single sentence can be processed through multiple levels of Natural Language Processing (NLP).